Traits in Farm Animals

Classes of traits

- Traits or characters are terms applied to the features in which man has an interest with respect to his animals
- **Five basic categories**

A. Fitness traits*:* - Are usually concerned with reproduction and viability

- Ex.: Litter size, conception rate, calving interval, gestation length, survival ability

B. Production traits

- *Ex.: as milk yield, growth rate, feed efficiency, weaning weight*
  - ❖Need to be defined as to what part of the time scale is involved

C. Quality traits

- ➢ *Ex.: Carcass composition, back-fat thickness in pig/sheep, eye muscle area in all carcass, meat quality, milk-fat percentage*

D. Type traits

- Ex.: *coat colour, coat type, udder shape, physical appearance, horn shape, etc.*

E. Behavioral traits

- ▪ *Ex.: herding ability in sheepdogs, docility, response to stress.*
- *Important in so far as animal welfare is concerned*

Measurement of traits

- Traits can be classified as being either objective or subjective
- Most traits can be measured, e.g. weight gain, milk yield, fleece weight, litter size
- Some are subjectively assessed using a scoring system or a scale
  - to type as well as to some carcass characteristics
- Breeding is frequently about making comparisons b/n animals
- Thus important that measurements are made on valid terms
- It is, for example, difficult and unwise to compare animals reared and managed in different herds or to compare animals of different ages

Measurement of traits Cont . . .

- Adjustment or correction factors have usually to be applied before some data can be analyzed and comparisons made
- Some of the factors, which need to be considered before comparisons include:
- Herd/year/season
- Age of animal
- Sex of the animal
- Litter size
- Dam age
- Parity

Traits of importance

- Traits vary from species to species, they also vary in importance at different periods of time
- *For example,*
  - *If the milk producer derives income largely from yield then total milk production will be more important than milk composition*
  - *A pig breeder is interested in litter size and litters produced per year but the buyer of weaned pigs has no interest in this*

Economically important traits of dairy cattle

Reproductive

- Age at first service
- Age at first conception
- Age at first calving
- Service per conception
- Service period/ longevity
- Calving interval
- Twinning
- Dystocia

Economically important traits of dairy cattle

Yield per 305 day lactation

- Milk yield
- Milk fat
- Protein
- Solids-not-fat

Other features

- Somatic cell count (as indicator of mastitis)
- Rate of milk late down
- Temperament

Economically important traits of beef cattle

Reproduction

- Age at puberty/first mating
- Number of live births
- Service per conception
- Dystocia

Maternal ability

- Weight calf weaned/cow
- Mothering ability
- Temperament

Economically important traits of beef cattle

- **Carcass traits**
  - Killing out (dressing) percent
  - Fat depth (various locations)
  - Longissimus dorsi area
  - Classification/ grade
  - Weight of saleable meat
  - Weight of excess fat
  - Weight of bone
  - Proportion of lean/fat/bone
  - Ratio meat: bone
  - Muscling/shape

Economically important traits of beef cattle

Weight

- Birth
- Weaning
- Yearling
- Pre-weaning gain
- Post-weaning gain
- 400-day weight

Other traits

- Type aspects
- Disease resistance
- Heat resistance (in tropics)

Economically important traits of sheep

- **Reproduction**
  - Age at first mating
  - Number born
  - Litters per year
- **Maternal ability**
  - Milk production
  - Temperament
  - Survival ability of lamb
- **Carcass**
  - Dressing %
  - Fat depth
  - Proportions (meat, fat, bone)
  - Longissimus dorsi area

Economically important traits of sheep

- **Weight**
  - Birth
  - Weaning
  - Slaughter
  - Pre-weaning gain
  - Post-weaning gain
  - Age at slaughter
- **Fleece**
  - Fleece Weight
  - Fiber diameter
  - Staple length (wool length)
  - Color
- **Others**
  - Type and breed characteristics